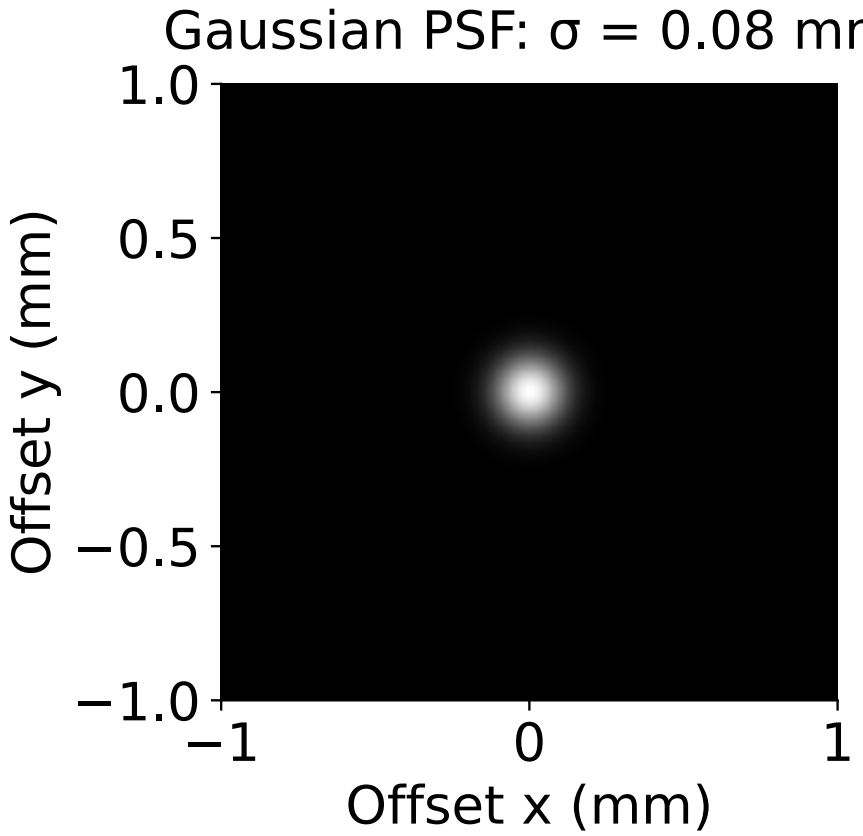
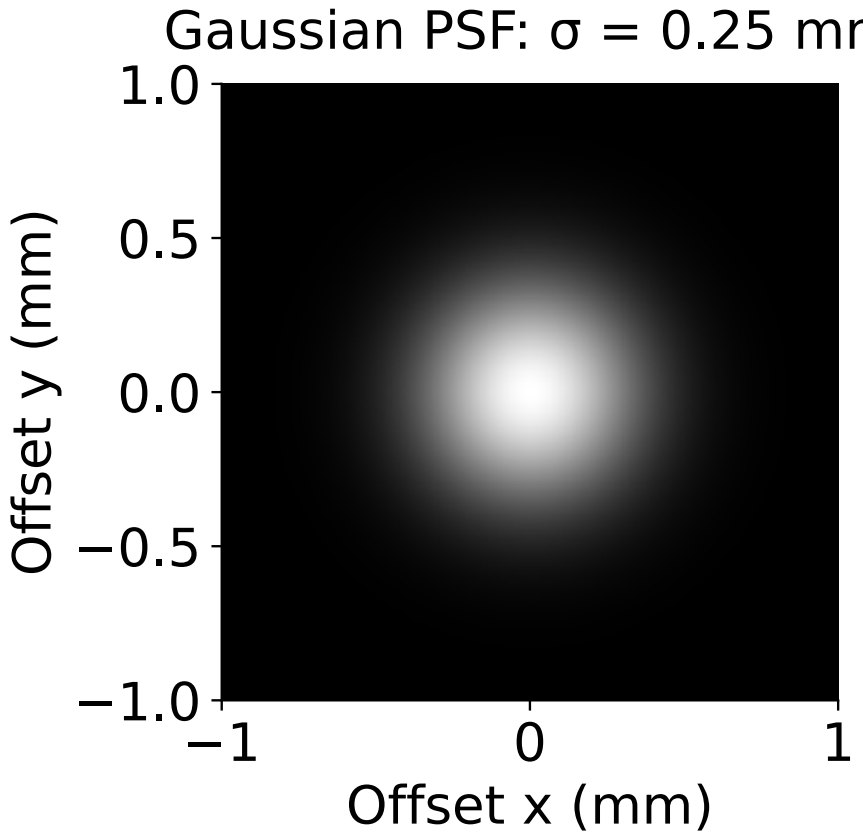
System response

A point becomes a spot





The **point-spread function (PSF)** describes the image of an ideal point input.

Linear, shift-invariant imaging

For a **linear** system, weighted input signals produce the same weighted combination of output signals.

For a **shift-invariant** system, moving an input point shifts its response without changing its shape.

These assumptions allow one PSF to describe the system over the region of interest.

Convolution and image formation

$$I_{\text{blur}}(x, y) = (I_0 * h)(x, y).$$

The system spreads each object point according to the PSF and adds the contributions.

$$I_{\text{blur}}(x, y) = \iint I_0(\xi, \eta)\, h(x - \xi, y - \eta)\, d\xi\, d\eta.$$

For an intensity-preserving blur model, $\iint h(x, y)\, dx\, dy = 1$.

Object and image in the frequency domain

Use $f(x, y)$ for the object, $h(x, y)$ for the impulse response, and $g(x, y)$ for the output image.

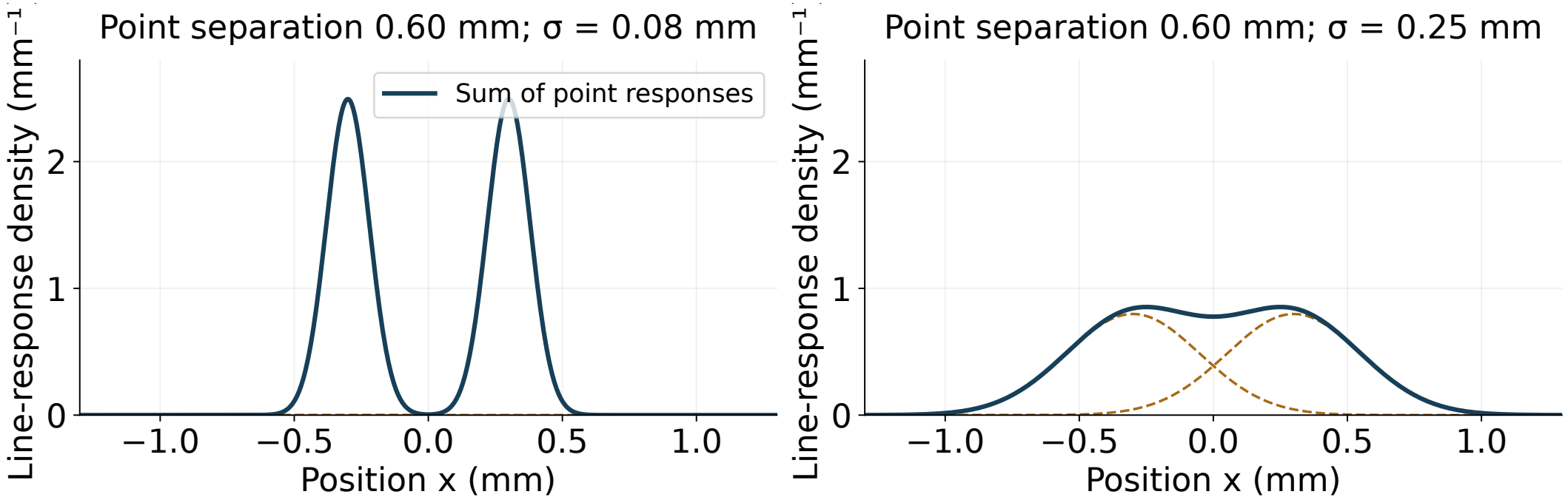
$$g = f * h \quad \Longleftrightarrow \quad G(u, v) = F(u, v)H(u, v).$$

The **transfer function** H is the Fourier transform of h. Each object's frequency component is multiplied by the system's response at that frequency.

A constant input probes the zero-frequency response:

$$H(0, 0) = \iint h(x, y)\, dx\, dy.$$

Two nearby point inputs



A wider PSF makes the two responses overlap more strongly.

Gaussian PSF width

$$h(x,y) = \frac{1}{2\pi\sigma^2} \exp\left[-\frac{x^2+y^2}{2\sigma^2}\right].$$

σ sets the width of the spot and has units of mm.

The **full width at half maximum (FWHM)** is the width across the central profile at half its peak value:

$$\text{FWHM} = 2\sqrt{2\ln 2}\,\sigma \approx 2.355\sigma.$$

Example: $\sigma = 0.10$ mm gives FWHM ≈ 0.236 mm.

Separable Gaussian responses

If $h(x, y) = h_x(x)h_y(y)$, its transform factors:

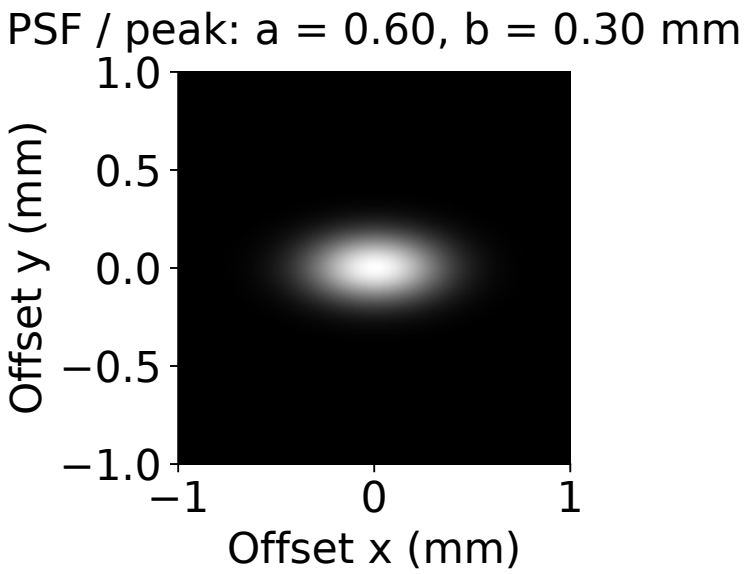
$$H(u, v) = H_x(u)H_y(v).$$

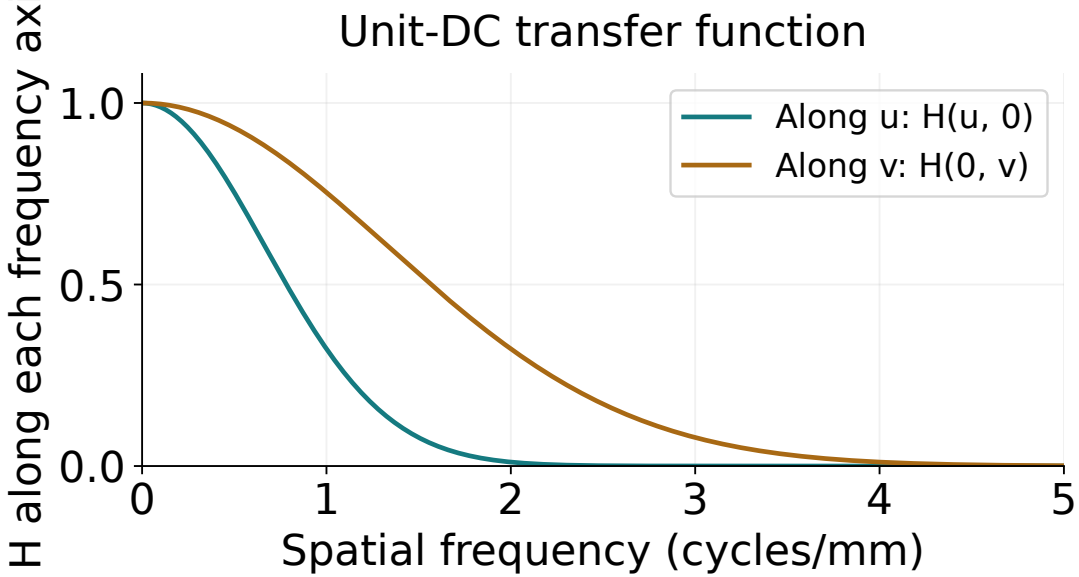
For $a, b > 0$,

$$h(x, y) = Ae^{-\pi[(x/a)^2+(y/b)^2]} \quad \Longrightarrow \quad H(u, v) = Aab\, e^{-\pi[(au)^2+(bv)^2]}.$$

Method: transform each one-dimensional Gaussian, include its scale factor, and multiply. Check $H(0, 0)$ against the integral of h.

Resolution can differ by direction





In this example, the PSF is wider along x. The response falls more rapidly with u, so the system preserves finer detail along y.

Sinusoidal contrast

$$C = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}.$$

For $I(x) = \bar{I}[1 + C\cos(2\pi f x + \phi)]$, contrast is the amplitude divided by the mean.

The notebook target has $\bar{I} = 0.5$ and $C_0 = 0.9$, giving $I_{\max} = 0.95$ and $I_{\min} = 0.05$.

Blur can reduce contrast without changing the stripe frequency.

Modulation transfer function

The **MTF** describes how much sinusoidal contrast the system preserves at each spatial frequency:

$$\mathrm{MTF}(f) = \frac{C_{\mathrm{out}}(f)}{C_{\mathrm{in}}(f)}.$$

For our normalized LSI blur model, it is the magnitude of the Fourier transform of the PSF.

An MTF of 0.5 at a particular frequency means the output sinusoid has half the input contrast at that frequency.

MTF normalization and DC gain

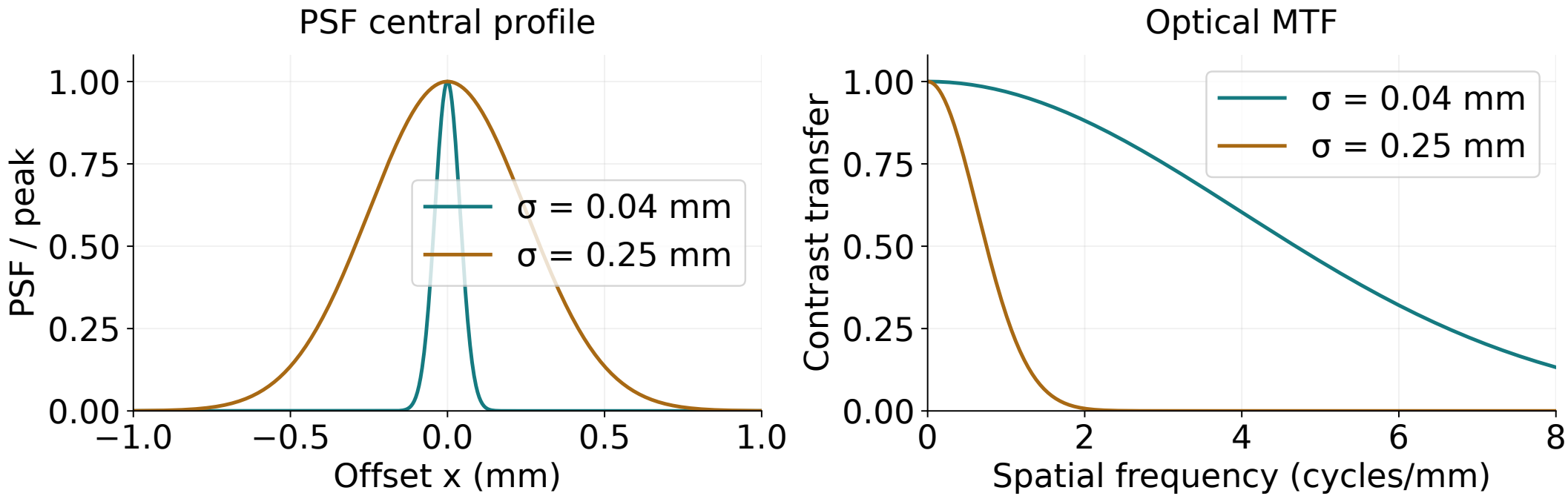
For an LSI system with positive DC gain,

$$\text{MTF}(u, v) = \frac{|H(u, v)|}{|H(0, 0)|}.$$

Dividing by the zero-frequency gain compares sinusoidal amplitude transfer with mean-intensity transfer. This gives contrast transfer and makes $\text{MTF}(0, 0) = 1$.

Example: if $H(0, 0) = 4$ and $|H(u_1, v_1)| = 2$, the MTF there is 0.5. An input contrast of 0.8 becomes 0.4.

PSF width and contrast transfer



For a Gaussian, $H_{\text{opt}}(f) = \exp(-2\pi^2\sigma^2 f^2)$.

Finite pixel size

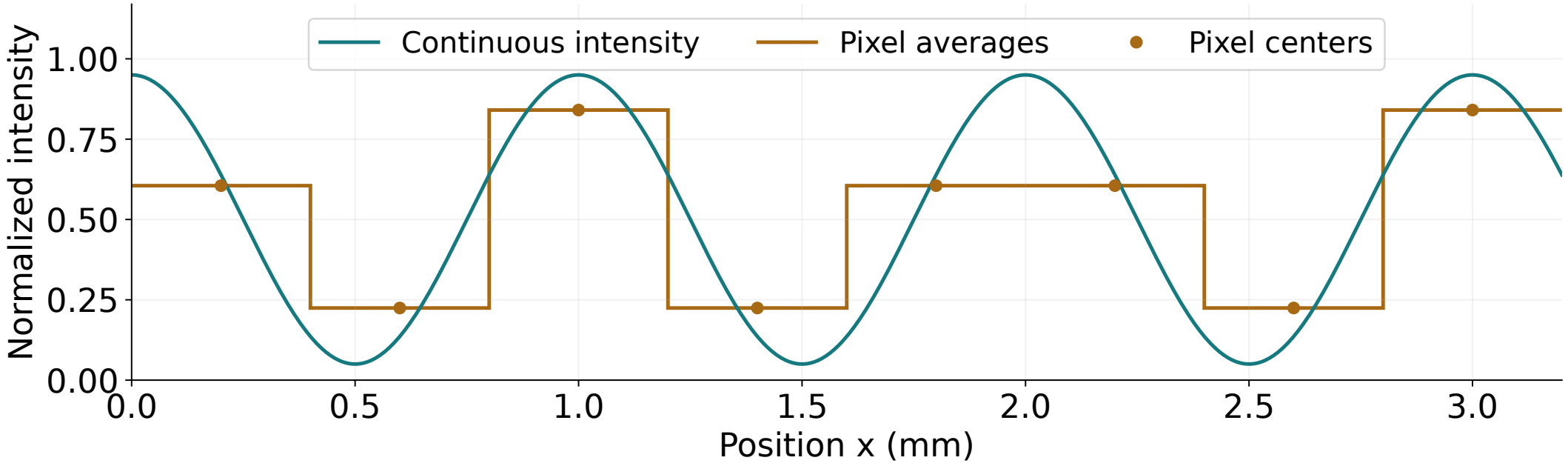
A pixel reports an average over its collecting area, followed by one value at its center.

For vertical stripes and pixels of width and spacing p,

$$d_n = \frac{1}{p}\int_{x_n - p/2}^{x_n + p/2} I_{\text{blur}}(x)\,dx, \qquad x_n = (n + \tfrac{1}{2})p.$$

Larger pixels average over more of the spatial variation.

Averaging and sampling



The step display assigns each recorded average to its pixel area.

Combined response before sampling

For a sinusoid at frequency f, pixel averaging contributes

$$H_{\text{pixel}}(f) = \text{sinc}(pf), \qquad \text{sinc}(u) = \frac{\sin(\pi u)}{\pi u}.$$

The combined **presampling MTF** is

$$\text{MTF}_{\text{pre}}(f) = |H_{\text{opt}}(f) H_{\text{pixel}}(f)|.$$

Sampling adds a separate limit: $f_N = 1/(2p)$.

Contrast loss below Nyquist

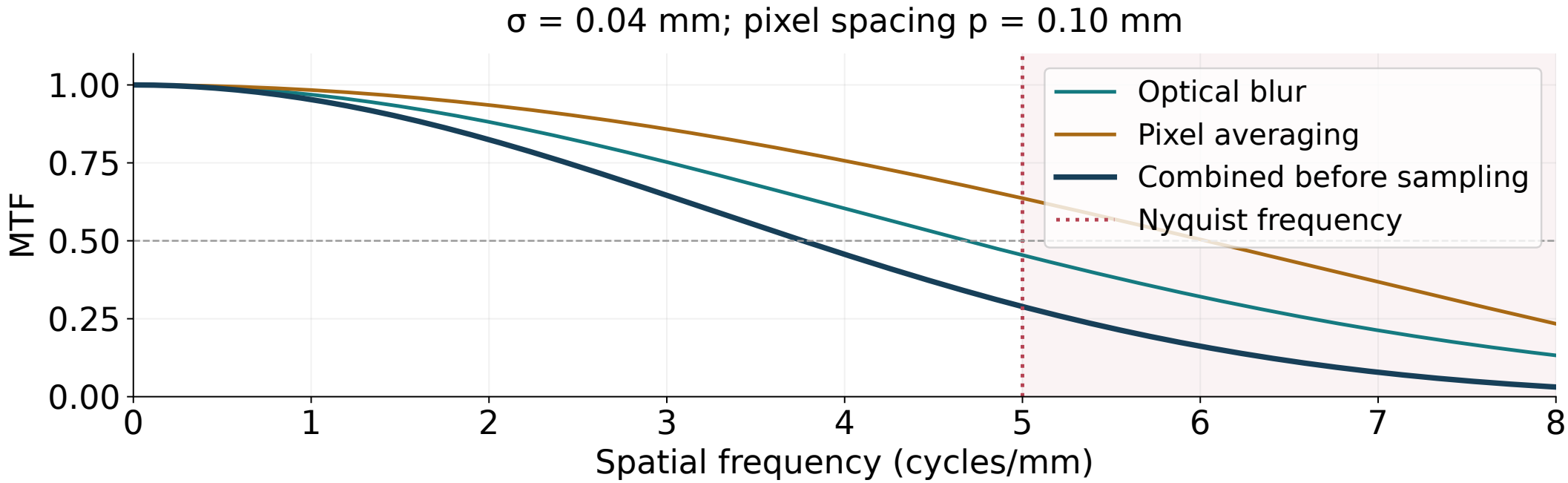
Take a sinusoid at $f = 1$ cycle/mm and pixels of width and spacing $p = 0.40$ mm.

$$f_N = \frac{1}{2p} = 1.25 \text{ cycles/mm}, \qquad |H_{\text{pixel}}(f)| = |\operatorname{sinc}(0.40)| \approx 0.757.$$

The frequency is below Nyquist, yet pixel averaging preserves only about 76% of the continuous sinusoidal contrast.

- **Averaging:** nearby bright and dark intensities contribute to one pixel value.
- **Aliasing:** different continuous frequencies produce the same samples.

An MTF curve and a sampling limit



MTF50 is the first frequency where the combined MTF falls to 0.5.

Guided notebook investigation

Open **01_sampling_resolution.ipynb** and run all cells.

1. Select **Fine sampling**. Identify the object, optical image, and detector image.
2. Predict what will change when the PSF gets wider.
3. Select **Blur limited**. Keep frequency and pixel spacing fixed.
4. Compare the images, intensity profile, and MTF curve.

Explain the observed change to a partner before recording your answer.

Measurements for the comparison

Quantity	Fine sampling	Blur limited
PSF width σ (mm)	0.04	0.25
FWHM $\approx 2.355\sigma$ (mm)	Calculate	Calculate
MTF at 1 cycle/mm	Read	Read
MTF50 (cycles/mm)	Read	Read

Why does stripe contrast decrease even though there are still ten pixels per stripe period?

Exit questions

1. What does the PSF describe, and what does the MTF describe?
2. If $C_{\text{in}} = 0.8$ and MTF $= 0.5$, what is C_{out}?
3. What can limit spatial detail even when the pixel spacing is small?

Next meeting: image quality, detector sampling, and aliasing in the notebook.

Reading and resources

Prince and Links, *Medical Imaging: Signals and Systems*: LSI imaging, point-spread functions, and image quality.

- Michigan EECS 516: Medical Imaging Systems
- MIT RES.6-007: convolution and Fourier-transform review
- Course notebook: **01_sampling_resolution.ipynb**, investigation 1.